Intrumental Variables

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Motivation

Three important threats to internal validity:

- Omitted variables (Unobserved, so cannot be included in the regression) – If not constant over time or states, panel data do not help
- Measurement error (X is measured with error)
- Simultaneity (X causes Y, Y causes X)

All of them imply that $E[u|X] \neq 0$, and OLS estimator is biased and inconsistent.

Instrumental variables (IV) regression can eliminate bias from these three sources.

Measurement Error

- Data is often measured with error:
  - reporting errors.
  - coding errors.
- When the measurement error is in the dependent variable, the zero conditional mean assumption is not violated and thus no endogeneity.
- In contrast, when the measure error is in the independent variable, the problem of endogeneity arises.

Measurement Error

- Consider a simple regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i.$$

- X_i is measured with errors. That is, we observe $\widetilde{X}_i = X_i + e_i$ instead of X_i.
- We assume that e_i is uncorrelated with X_i, $E(X_i e_i) = 0$.
- Then, the regression equation we use is

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 \widetilde{X}_i + u_i - \beta_1 e_i \\ &= \beta_0 + \beta_1 \widetilde{X}_i + v_i. \end{aligned}$$

- It can be seen that the problem of endogeneity occurs:

$$\begin{aligned} E\left(\widetilde{X}_i v_i\right) &= E\left(\left(X_i + e_i\right)\left(u_i - \beta_1 e_i\right)\right) \\ &= -\beta_1 Var\left(e_i\right) \neq 0. \end{aligned}$$

Simultaneity (Reverse Causality)

- **Simultaneity** arises when one or more of the independent variables, X_js, is jointly determined with the dependent variable, Y, typically through an equilibrium mechanism.
- This arises in many economic contexts:
  - quantity and price by demand and supply
  - investment and productivity
  - sales and advertizement

Simultaneity

- Suppose that the equilibrium relation between X and Y is expressed by the following simultaneous equations:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + u_i, \\ X_i &= \alpha_0 + \alpha_1 Y_i + v_i. \end{aligned}$$

- Each one is called a *structural equation* since it has a ceteris paribus, causal interpretation.
- By solving two equations, we have

$$\begin{aligned} Y_i &= \frac{\beta_0 + \beta_1 \alpha_0}{1 - \alpha_1 \beta_1} + \frac{\beta_1 v_i + u_i}{1 - \alpha_1 \beta_1}, \\ X_i &= \frac{\alpha_0 + \alpha_1 \beta_0}{1 - \alpha_1 \beta_1} + \frac{v_i + \alpha_1 u_i}{1 - \alpha_1 \beta_1}. \end{aligned}$$

Simultaneity

- These are called the *reduced form* of the model. It is easy to see that, if one perform the regression with just one equation, it will lead to a biased OLS estimator, called **simultaneity bias**.

$$\begin{aligned} Cov\left(X_i, u_i\right) &= Cov\left(\frac{v_i + \alpha_1 u_i}{1 - \alpha_1 \beta_1}, u_i\right) \\ &= \frac{\alpha_1}{1 - \alpha_1 \beta_1} Var\left(u_i\right). \end{aligned}$$

What Is IV?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- Given that X is correlated with u, we can break X into two parts: a part that might be correlated with u, and a part that is not. By isolating the part that is not correlated with u, it is possible to estimate β_1
- This can be done by an instrumental variable (IV), denoted as Z, which is uncorrelated with u
- The IV Z detects movements in X that are uncorrelated with u, and thus estimate β_1

What Is IV?

Suppose Z is uncorrelated with u, and X is correlated with u. How to do the decomposition?

- Write

$$X = \pi_0 + \pi_1 Z + v \, .$$

  We need to find π_0 and π_1.
- We can use two restrictions: $E[v] = 0$ and $Cov(v, Z) = 0$. This two restrictions imply that "$\pi_0 + \pi_1 Z$" and "v" are uncorrelated as well.
- Solving the two restrictions give

$$\pi_1 = \frac{Cov(Z, X)}{Var(Z)} \text{ and } \pi_0 = E[X] - \pi_1 E[Z] \, .$$

Endogeneity v.s. Exogeneity

- An endogenous variable is one that is correlated with u – Threats to interval validity imply endogeneity of X
- An exogenous variable is one that is uncorrelated with u – IV Z is exogenous

So, IV regression uses exogenous IV Z as tool to estimate the causal effect β_1 of endogenous regressor X.

Historical note: "Endogenous" literally means "determined within the system," that is, a variable that is jointly determined with Y, that is, a variable that is subject to simultaneous causality. However, this definition is narrow and IV regression can be used to address OVB and errors-in-variable bias, not just to simultaneous causality bias.

Two Conditions: Validity of IV

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

For an instrumental variable (an "instrument") Z to be valid, it must satisfy two conditions:

1. Relevance: $corr(Z, X) \neq 0$
2. Exogeneity: $corr(Z, u) = 0$

Alternatively, you may explain: Z affects Y only through X.

- IV is not simply a proxy for the causal variable of interest
- IV is not an omitted variable

Suppose for now that you have such a Z (we'll discuss how to find instrumental variables later). How can you use Z to estimate β_1?

Two Stage Least Squares (2SLS)

As it sounds, 2SLS has two stages – two regressions:

1. First, isolate the part of X that is uncorrelated with u: regress X on Z using OLS

$$X_i = \pi_0 + \pi_1 Z_i + v_i \tag{1}$$

   - Because Z_i is uncorrelated with u_i, $\pi_0 + \pi_1 Z_i$ is uncorrelated with u_i. We don't know π_0 or π_1 (population parameters) but we have estimated them, so ...
   - Compute the predicted values of X_i, $\hat{X}_i$, where $\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i, i = 1, \ldots, n$.

Two Stage Least Squares (2SLS)

2. Replace X_i by $\hat{X}_i$ in the regression of interest: regress Y on $\hat{X}_i$ using OLS:

$$Y_i = \beta_0 + \beta_1 \hat{X}_i + u_i \tag{2}$$

- Because $\hat{X}_i$ is uncorrelated with u_i in large samples, so the first least squares assumption holds
- Thus, β_1 can be estimated by OLS using regression (2)
- This argument relies on large samples (so π_0 and π_1 are well estimated using regression (1))
- This the resulting estimator is called the "Two Stage Least Squares" (2SLS) estimator. It is also called IV estimator

Two Stage Least Squares (2SLS)

By the OLS formula,

$$\hat{\beta}_{1,IV} = \frac{n^{-1}\sum_{i=1}^{n}(\hat{X}_i - \bar{\hat{X}})(Y_i - \bar{Y})}{n^{-1}\sum_{i=1}^{n}(\hat{X}_i - \bar{\hat{X}})^2}$$

$$= \frac{\hat{cov}(\hat{X}, Y)}{\hat{var}(\hat{X})} \text{ (notation for sample covariance)}$$

$$= \frac{\hat{cov}(\hat{\pi}_0 + \hat{\pi}_1 Z, Y)}{\hat{var}(\hat{\pi}_0 + \hat{\pi}_1 Z)} \text{ (plug in } \hat{X}\text{)}$$

$$= \frac{\hat{\pi}_1 \hat{cov}(Z, Y)}{\hat{\pi}_1^2 \hat{var}(Z)} \text{ (by the properties of } \hat{cov}\text{)}$$

$$= \frac{\hat{cov}(Z, Y)}{\hat{\pi}_1 \hat{var}(Z)} = \frac{\hat{cov}(Z, Y)}{\hat{cov}(Z, X)} \text{ (plug in } \hat{\pi}_1\text{)}$$

$$= \frac{n^{-1}\sum_{i=1}^{n}(Z_i - \bar{Z})(Y_i - \bar{Y})}{n^{-1}\sum_{i=1}^{n}(Z_i - \bar{Z})(X_i - \bar{X})}.$$

Two Stage Least Squares (2SLS)

Alternative explanation: (only) a little algebra

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- From the exogeneity of Z ($cov(Z, u) = 0$), we have

$$\begin{aligned} 0 = cov(Z, u) &= cov(Z, Y - \beta_0 - \beta_1 X) \\ &= cov(Z, Y) - \beta_1 cov(Z, X) \end{aligned}$$

  - Exogeneity condition is the information you have

- From the relevance of Z ($cov(Z, X) \neq 0$), we have

$$\beta_1 = \frac{cov(Z, Y)}{cov(Z, X)}$$

  - Relevance condition ensures that we do not divide by zero

Two Stage Least Squares (2SLS)

$$\beta_1 = \frac{cov(Z, Y)}{cov(Z, X)}$$

- β_1 is equal to the ratio of two covariances
- Thus, we can estimate β_1 by replacing these population covariances with sample covariances – this is the method of moment estimator
- The resulting estimator is identical to $\hat{\beta}_{1,IV}$, so the IV estimator can be explained as the method of moment estimator

Note: If $Z = X$, the IV estimator becomes the OLS estimator!

How to Find IV?

Omitted variables:

$$Y_i = \beta_0 + \beta_1 X_i + \gamma W_i + v \, .$$

- Use your judgement and theory: the IV Z satisfies the two conditions or affects Y only through X
- IV is not the omitted variable: $Z \neq W$

Measurement error:

- When the economist is worried about measurement error, a good choice of instrument is simply a different measure of the same variable
- The new measure may have its own errors, but these errors are unlikely to be correlated with the mistakes in the first measure, or with any other component of u

How to Find IV?

Simultaneity:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$
$$X_i = \gamma_0 + \gamma_1 Y_i + \gamma_2 Z_i + v_i$$

- Use Z that only appears in the second equation as an IV to estimate the first equation (β_1)
- See the math illustration in class in the estimation of supply and demand equations

How to Find IV?

- Levitt (1997): effect of police on crime
  - Concern: high-crime areas require more police
  - IV: Mayor electoral cycle - affect the number of police without affecting crime
- Angrist and Krueger (1991): effect of schooling on future income
  - Concern: unobserved ability affects both schooling and income
  - IV: quarter of birth - affect years of schooling without affecting ability
- Angrist and Evans (1998): impact of childbirth on women labor participation
  - Concern: women with low earning potential have more kids and do not work
  - IV: gender mix of the first two kids - affect the number of children without affecting earning potential

Example: Demand for Cigarettes

- Suppose you are studying the effect of price on the demand for cigarettes, using the data of different states' cigarette consumption and average price
- To answer this, we need to estimate the elasticity of demand for cigarettes, that is, β_1, in the regression,

$$\ln(Q_i^{cigarettes}) = \beta_0 + \beta_1 \ln(P_i^{cigarettes}) + u_i$$

- Note that we sometimes omit the intercept to simplify the presentation but in practice an intercept should always be included
- Will the OLS estimator plausibly be unbiased? *Why or why not?*

Example: Demand for Cigarettes

Panel data:

- Annual cigarette consumption and average prices paid (including tax)
- 48 continental US states, 1985-1995

Proposed instrumental variable:

- Z_i = general sales tax per pack in the state = $SalesTax_i$
- Is this a valid instrument?
  1. Relevant? $corr(SalesTax_i, \ln(P_i^{cigarettes})) \neq 0$?
  2. Exogenous? $corr(SalesTax_i, u_i) = 0$?

Example: Demand for Cigarettes

First stage regression:

```
reg lravgprs rtaxso if year==1995, r

Linear regression                               Number of obs =      48
                                                F(  1,    46) =   40.39
                                                Prob > F      =  0.0000
                                                R-squared     =  0.4710
                                                Root MSE      =  .09394

------------------------------------------------------------------------------
             |               Robust
    lravgprs |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      rtaxso |   .0307289   .0048354     6.35   0.000     .0209956    .0404621
       _cons |   4.616546   .0289177   159.64   0.000     4.558338    4.674755
------------------------------------------------------------------------------
. predict lravphat
```

$$\widehat{\ln(P^{cigarettes})} = \underset{(0.04)}{4.63} + \underset{(0.004)}{0.031} \times SaleTax, n = 48$$

Example: Demand for Cigarettes

Second stage regression:

```
reg lpackpc lravphat if year==1995, r

Linear regression                               Number of obs =      48
                                                F(  1,    46) =   10.54
                                                Prob > F      =  0.0022
                                                R-squared     =  0.1525
                                                Root MSE      =  .22645

------------------------------------------------------------------------------
             |               Robust
     lpackpc |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    lravphat |  -1.083586   .3336949    -3.25   0.002    -1.755279   -.4118932
       _cons |   9.719875   1.597119     6.09   0.000     6.505042    12.93471
------------------------------------------------------------------------------
```

$$\widehat{\ln(Q^{cigarettes})} = \underset{(1.60)}{9.72} - \underset{(0.33)}{1.08} \times \widehat{\ln(P^{cigarettes})}, n = 48$$

- These coefficients are the IV estimates
- The standard errors are wrong because they ignore the fact that the first stage was estimated

Example: Demand for Cigarettes

Use a single specialized command:

```
ivreg lpackpc (lravgprs = rtaxso) if year==1995, r

Instrumental variables (2SLS) regression          Number of obs =      48
                                                  F(  1,    46) =   11.54
                                                  Prob > F      =  0.0014
                                                  R-squared     =       .
                                                  Root MSE      =  .19035

------------------------------------------------------------------------------
             |               Robust
     lpackpc |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    lravgprs |  -1.083587   .3189183    -3.40   0.001    -1.725536   -.4416373
       _cons |   9.719876   1.528322     6.36   0.000     6.643525    12.79623
------------------------------------------------------------------------------
Instrumented:  lravgprs
Instruments:   rtaxso
------------------------------------------------------------------------------
```

$$\widehat{\ln(Q^{cigarettes})} = \underset{(1.52)}{9.72} - \underset{(0.32)}{1.08} \times \ln(P^{cigarettes}), n = 48$$

- OK, the change in the SEs was small this time . . . but not always!

Summary

- A valid instrument Z must satisfy two conditions:
  1. Relevance: $corr(Z, X) \neq 0$
  2. Exogeneity: $corr(Z, u) = 0$
- 2SLS proceeds by first regressing X on Z to get $\hat{X}$, then regressing Y on $\hat{X}$
- The key idea is that the first stage isolates part of the variation in X that is uncorrelated with u
- If the instrument is valid, then the large-sample sampling distribution of the 2SLS estimator is normal, so inference proceeds as usual